

GR00T N1 · arXiv 2503.14734 · Oct 7 club

Visionary

11+3 · HIGH · seat empty

N1.7 official BEHAVIOR'26 substrate
Dual-track vs $\pi_{0.5}$ from day one

Spine: Eagle-2 System 2 (~ 10 Hz) + FM DiT System 1 (120 Hz) via cross-attn

H=16 · K=4 · N1 $\neq$ N1.7 · NOT a classical planner

NVIDIA · Fourier GR-1 · open humanoid VLA

Visionary agenda · six pack bullets (MUST HIT)

HIGH for BEHAVIOR'26 · densest deck · no Discord / Drive

1

N1.7 official substrate

Dual-track vs $\pi_{0.5}$ from day one — pick by ablations

2

Explicit System-2 interface

Hierarchy / safety / stage hooks vs fused π expert

3

Port neural pyramid + RTC

Correlated-noise / soft-inpaint hybrids from π forks

4

Version discipline

Paper $H=16$ vs N1.7 $H \leq 40$ — don't mix APIs

5

Try both starters

Report NFE/horizon · with/without System-2 mid-layer

6

Durable idea

Modular VL $\leftrightarrow$ motor + open humanoid data pyramid

Lineage · N1 → Isaac → N1.6 → N1.7 → BEHAVIOR'26

CLEAR: N1.7 GA = official co-baseline with $\pi_{0.5}$ · do NOT mix APIs/horizons



VERSION DISCIPLINE

Paper N1 $\neq$ N1.7 (backbone / horizon / package)
N1.6 BEHAVIOR1k numbers $\neq$ N1.7 forecasts

π track (parallel)

$\pi_0 \rightarrow$ OpenPI $\rightarrow \pi_{0.5} +$ RTC
BEHAVIOR'25 forks $\rightarrow$ '26 co-baseline

Genealogy: SigLIP-2+SmolLM2 $\rightarrow$ Eagle-2 $\rightarrow$ N1 $\rightarrow$ N1.6 $\rightarrow$ N1.7 $\rightarrow$ BEHAVIOR'26 || $\pi_{0.5}$

GR00T N1 vs π_0 / $\pi_{0.5}$ · same CFM family, different handoff

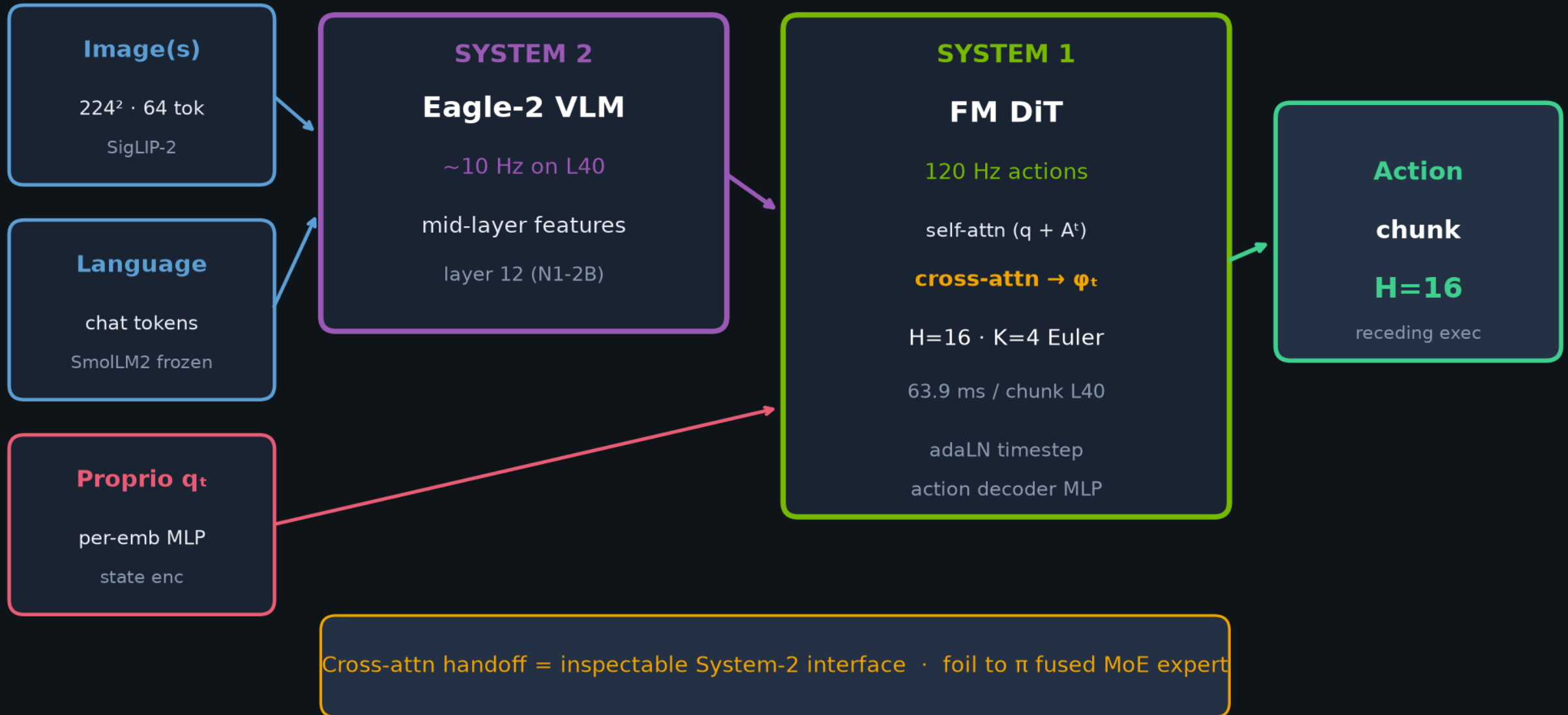
BEHAVIOR'26 ships BOTH as official baselines — dual-track is first-class

AXIS	GR00T N1	π_0 / $\pi_{0.5}$
Coupling	Separate DiT · cross-attn	Action expert · shared self-attn
VLM	Eagle-2 (SmolLM2+SigLIP-2)	PaliGemma lineage
Chunk / denoise	H=16 · K=4 · 10/120 Hz	H=50 (π_0) · ~10 Euler · 20-50 Hz
Embodiment I/O	Per-emb MLPs · LAPA/IDM	Pad/mask shared dim
Data bet	Pyramid + neural I2V	OXE-scale + $\pi_{0.5}$ co-train
Successor	N1.7 ~3B · H≤40 · Apache-2.0	$\pi_{0.5}$ + RTC + BEHAVIOR forks

FLAG: modular dual-system / cross-attn DiT vs fused shared-attn MoE expert · same chunk+flow interface

Dual-system VLA · Eagle-2 System 2 × FM DiT System 1

Jointly trained end-to-end · cross-attn bridge · NOT a classical planner



Data pyramid · quantity ↓ · embodiment-specificity ↑

~593M frames / ~8376 h · human + neural + DexMimicGen + real

PEAK · Real robot

GR-1 teleop ~88 h · OXE · AgiBot-Alpha · RH20T

MIDDLE · Synthetic

DexMimicGen ~780k traj · neural I2V 88h→827h · LAPA/IDM

BASE · Web + human egocentric

Ego4D · Ego-Exo4D · EPIC · HOI4D · Assembly-101 · LAPA latent actions

Visionary durable idea: open humanoid data pyramid survives backbone swaps (Eagle → Cosmos-Reason2 → ...)

VERSION DISCIPLINE · N1 ≠ N1.7

APIs / horizons / backbones differ — pin the checkpoint you serve

Paper N1

GR00T-N1-2B · ~2.2B (1.34B VLM)

Eagle-2 · SigLIP-2 + SmolLM2

H=16 · K=4 · mid-layer 12

63.9 ms / chunk @ L40 bf16

Table 2/3 tattoo numbers live [HERE](#)

N1.7 GA

nvidia/GR00T-N1.7-3B · ~3B

Cosmos-Reason2-2B (Qwen3-VL)

H ≤ 40 · new package / API

Apache-2.0 · BEHAVIOR'26 baseline

Do NOT quote N1 tables as N1.7

Stakeholder / Visionary rule

Prefer finetune N1.7 for product/challenge · use N1-2B only for paper replication

Match served horizon to trained horizon · N1.6 BEHAVIOR1k ≠ N1.7 forecast

Also: N1 is NOT a classical planner — generative CFM chunk controller with dual-system packaging

What to try for 2026 · grounded bets

Organizer metric > paper tabletop % · released weights + FT, not re-pretrain

Dual bring-up

Both official baselines on one hard task
(turning_on_radio starters exist)

Horizon / NFE

H 16 vs 40 if base allows
K sweep · log ms/chunk

System-2 mid-layer

With vs without mid-layer tap
stage memory / safety filter hooks

$\pi \rightarrow$ GR00T hybrids

RTC soft-inpaint · correlated noise
on N1.7 action head

Forgetting probes

Held-out skill probes during FT
mix pyramid replay

Compute politics

Student path = N1.7 + adapter FT
NOT 50k H100-h Eagle retrain

Durable idea · what survives backbone swaps

Modular VL ↔ motor coupling

+ open humanoid data pyramid

— not just a bigger PaliGemma clone

Handoff

Inspectable System-2
features for hierarchy

Projectors

Per-embodiment MLPs
multi-robot bring-up

Pyramid

Human · neural · sim · real
flywheel > clever head alone

Backbone will keep swapping (Eagle → Cosmos-Reason2 → ...); productize the interface

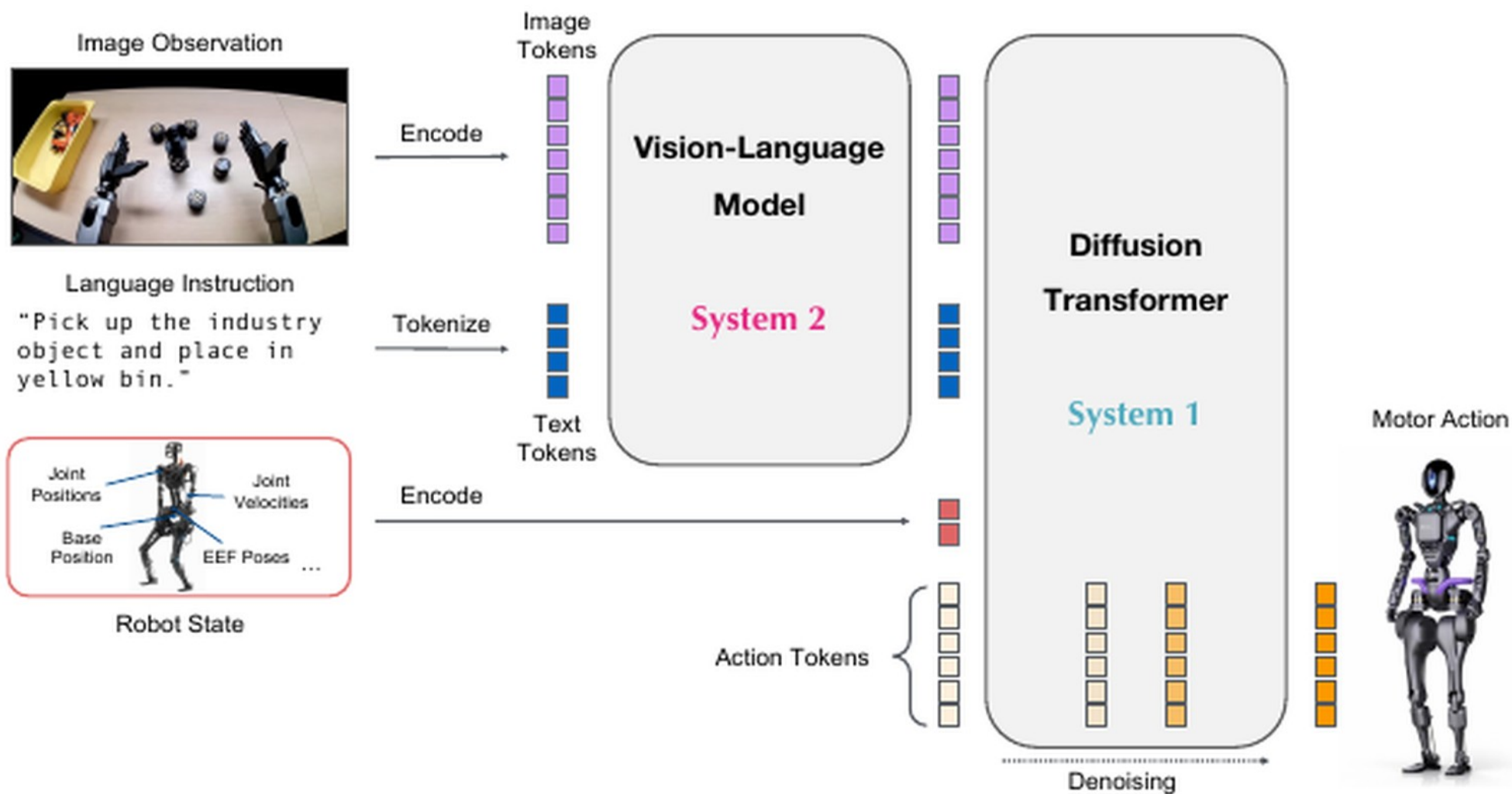


Figure 2: GR00T N1 Model Overview. Our model is a Vision Language Action (VLA) model that adopts a

control modes, and other factors result in an archipelago of “data islands” rather than a coherent, Internet-scale dataset needed for training a true generalist model.

We introduce GR00T N1, an open foundation model for generalist humanoid robots. The GR00T N1 model is a Vision-Language-Action (VLA) model, which generates actions from image and language instruction input. It has cross-embodiment support from tabletop robot arms to dexterous humanoid robots. It adopts a **dual-system compositional architecture**, inspired by human cognitive processing (Kahneman, 2011). The System 2 reasoning module is a pre-trained Vision-Language Model (VLM) that runs at 10Hz on an NVIDIA L40 GPU. It processes the robot’s visual perception and language instruction to interpret the environment and understand the task goal. Subsequently, a Diffusion Transformer, trained with action flow-matching, serves as the System 1 action module. It cross-attends to the VLM output tokens and employs embodiment-specific encoders and decoders to handle variable state and action dimensions for motion generation. It generates closed-loop motor actions at a higher frequency (120Hz). Both the System 1 and System 2 modules



Figure 1: Data Pyramid for Robot Foundation Model Training. GR00T N1’s heterogeneous training corpora can be represented as a pyramid: data quantity decreases, and embodiment-specificity increases, moving from the bottom to the top.

Visionary close

Durable idea: modular VL↔motor + open pyramid

Try both starters · report NFE/horizon · System-2 mid-layer
No Discord / Drive in this briefing

GR00T N1 · Oct 7 · all seats empty · still full five packs

Control spine · CFM cousin, dual-system packaging

NOT classical planner · generative continuous chunk controller

$$A_t^\tau = \tau A_t + (1 - \tau)\varepsilon \quad \cdot \quad \mathcal{L}_{fm} = \mathbb{E} \|V_\theta - (\varepsilon - A_t)\|^2$$

VL @ 10 Hz

Eagle-2 mid features
cache across denoise

Denoise K=4

Euler on DiT
63.9 ms / H=16

Commit + replan

Don't wait full H
open-loop; recede

Contrast: OpenPI action-expert FM · DP CNN/Transformer · π_{RL} PPO · Comet RFT

Same CFM math as π_0 — different reasoning↔motor handoff

GR00T N1: An Open Foundation Model for Generalist Humanoid Robots

NVIDIA¹

Abstract

General-purpose robots need a versatile body and an intelligent mind. Recent advancements in humanoid robots have shown great promise as a hardware platform for building generalist autonomy in the human world. A robot foundation model, trained on massive and diverse data sources, is essential for enabling the robots to reason about novel situations, robustly handle real-world variability, and rapidly learn new tasks. To this end, we introduce GR00T N1, an open foundation model for humanoid robots. GR00T N1 is a Vision-Language-Action (VLA) model with a dual-system architecture. The vision-language module (System 2) interprets the environment through vision and language instructions. The subsequent diffusion transformer module (System 1) generates fluid motor actions in real time. Both modules are tightly coupled and jointly trained end-to-end. We train GR00T N1 with a heterogeneous mixture of real-robot trajectories, human videos, and synthetically generated datasets. We show that our generalist robot model GR00T N1 outperforms the state-of-the-art imitation learning baselines on standard simulation benchmarks across multiple robot embodiments. Furthermore, we deploy our model on the Fourier GR-1